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Drill Problems 9

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COLUMBIA ACADEMY

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1. **System of Equations Solutions** (10 points)
How many solutions does the given system of equations have?

$$y = 2x - 5$$
$$y = 2x^2 - 18x + 45$$

A. 0
B. 1
C. 2
D. ∞

Answer:

□

2. **Ice Cream Sundae Toppings** (10 points)
The function $c(t) = 6.50 + 1.50t$ can be used to find the cost of an ice cream sundae, where t is the number of toppings added to the sundae. If Henry ordered an ice cream sundae, paid for it with \$20, and received \$6 in change, how many toppings did Henry put on his sundae?

A. 3
B. 5
C. 7
D. 9

Answer:

□

3. **Sugar in Soda** (10 points)
There are 3 teaspoons in a tablespoon. There are 4 tablespoons in $\frac{1}{4}$ cup. If there are 10 teaspoons of sugar in one can of soda, how many cups of sugar are in 12 cans of soda?

A. $\frac{5}{6}$
B. $2\frac{1}{2}$
C. 4
D. 10

Answer:

□

4. **Trigonometric Identity** (10 points)
Angle R in a right triangle measures $n°$. If $\sin n° = \frac{5}{13}$, what is $\cos(90 - n)°$?

Answer:

□

5. **Perpendicular Line y-Intercept** (10 points)
Line l is perpendicular to the line $-4x+2y=14$ and passes through the point $(4,6)$. What is the coordinate of the y-intercept of line l?

   A. $(0,4)$
   B. $(0,5)$
   C. $(0,7)$
   D. $(0,8)$

   Answer:

   □

6. **Bouquet of Flowers** (10 points)
Sandra is putting together a bouquet of flowers using tulips and roses. She can have a total of 24 flowers in the bouquet. Each tulip costs \$4 and each rose costs \$6. If she spends \$126 on the bouquet, how many roses does Sandra put in the bouquet?

   A. 9
   B. 12
   C. 15
   D. 18

   Answer:

   □

7. **Kinetic Energy Formula** (10 points)
The formula for finding the kinetic energy of an object is $K = \frac{1}{2}mv^2$, where K is the kinetic energy of the object in joules, m is the mass of the object in kilograms, and v is the velocity of the object in meters per second. Which equation correctly expresses v in terms of K and m?

   A. $v = \pm\sqrt{\frac{2K}{m}}$
   B. $v = \pm\sqrt{\frac{K}{2m}}$
   C. $v = \pm\sqrt{\frac{2m}{K}}$
   D. $v = \pm\sqrt{\frac{1}{2}Km}$

   Answer:

   □

8. **Quadratic Vertex Value** (10 points)
For a quadratic function $f(x) = x^2 + bx + c$, the value $f(3) = -4$ is the vertex of the function. What is the value of c?

Answer:

$\square$

9. **Pizza Price Scatterplot** (10 points)
The scatterplot shows the relationship between the diameter of pizza in inches at a restaurant and the price of the pizza per square inch in dollars.

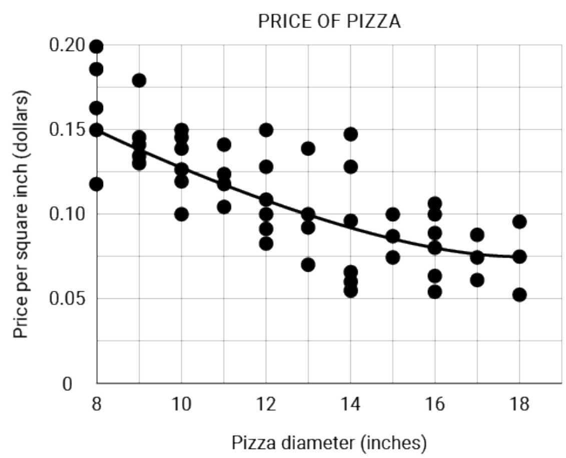

Figure 1: reference attached

Which of the following is the best estimate of the diameter, in inches, of a pizza that costs \$0.12 per square inch?

Answer:

$\square$

10. **Circle Center** (10 points)
The equation of the circle in the xy-plane is given:

$$x^2 + y^2 - 6x + 12y = 3$$

What is the center of the circle?

A. $(-6, 12)$
B. $(-3, 6)$
C. $(3, -6)$
D. $(6, -12)$

Answer:

$\square$

11. **Solving for x** (10 points)
If $4x + 16 = 24$, what is the value of $x + 4$?

A. 2
B. 4
C. 6
D. 8

Answer:

□

12. **Percent of a Value** (10 points)
If x is $2k\%$ of y, what is the value of $3k\%$ of $0.3y$?

A. $0.05x$
B. $0.2x$
C. $0.45x$
D. $0.9x$

Answer:

□

13. **Cubic Function Evaluation** (10 points)
For the given function $h(x) = 3x^3 - ax$, $h(2) = 20$. What is the value of $h(-1)$?

A. -5
B. -1
C. 1
D. 5

Answer:

□

14. **Trigonometric Relationship** (10 points)
An angle opposite a leg of a right triangle measures $w°$, and $\tan(w°) = \frac{3}{4}$. What is $\sin(90° - w°)$?

Answer:

□

15. **Exponent Rules** (10 points)
Which of the following is equivalent to $x^{0.6} \cdot x^{\frac{3}{20}}$ for all positive values of x?

   A. $\sqrt[20]{x^{0.9}}$
   B. $\sqrt[5]{x^3}$
   C. $\sqrt[4]{x^3}$
   D. $\sqrt[3]{x^4}$

   Answer:

   □

16. **System of Equations: Value of $a - b$** (10 points)

$$3.5a - 4.25b = -14.25$$
$$4.25a - 3.5b = -9$$

The solution to the given system of equations is (a, b). What is the value of $a - b$?

   A. -7
   B. -3
   C. 5
   D. 7

   Answer:

   □

17. **Mean vs. Median** (10 points)
A survey counted the number of flowers on the bushes in a park. The mean number of flowers per bush was 34 and the median number was 45. What factor most likely would account for the difference between the mean and the median?

   A. There were some bushes with extremely few flowers.
   B. The mode of the data was a number greater than 34.
   C. Many bushes had between 34 and 45 flowers.
   D. There is little variance in the number of flowers on the bushes.

   Answer:

   □

18. **Coffee Preferences Table** (10 points)
The table shows the distribution of coffee preferences among the 27 members of the DC Book Club. If a

	Cream	No Cream	Total
Sugar	4	7	11
No Sugar	15	1	16
Total	19	8	27

club member is selected at random, what is the probability the member prefers no cream, given that they prefer sugar?

A. $\frac{7}{27}$
B. $\frac{7}{16}$
C. $\frac{7}{11}$
D. $\frac{7}{8}$

Answer:

□

19. **Point on a Line** (10 points)
A line in the xy-plane contains the point $(3, -5)$ and has a slope of $\frac{4}{3}$. Which of the following points lies on the line?

A. $\left(\frac{1}{2}, 1\frac{2}{3}\right)$
B. $\left(1\frac{2}{3}, -5\right)$
C. $\left(2, -6\frac{1}{3}\right)$
D. $\left(5, -5\frac{1}{4}\right)$

Answer:

□

20. **Volume of a Cone** (10 points)
The height of a cone is three times its diameter. The base of the cone is a circle, and the area of the base is 9π square centimeters. What is the volume of the cone, in cubic centimeters?

A. 27π
B. 54π
C. 162π
D. 216π

Answer:

□

21. **Compound Interest Formula** (10 points)
A formula used for compound interest is $A = P\left(1+\frac{r}{n}\right)^{nt}$. Which of the following represents P in terms of A, n, and t?

A. $\frac{\left(1+\frac{r}{n}\right)^{nt}}{A}$
B. $A\left(1+\frac{r}{n}\right)^{nt}$
C. $\frac{A}{\left(1+\frac{r}{n}\right)^{nt}}$
D. $\frac{1}{A\left(1+\frac{r}{n}\right)^{nt}}$

Answer:

□

22. **Parking Garage Usage** (10 points)
A parking garage charges \$11 on weekdays and \$15 on weekends. If the garage brought in $\$53,234$ in one week and a total of 4,122 people used the garage, how many people used the garage during the weekend?

A. 304
B. 1,973
C. 2,149
D. 3,549

Answer:

□

23. **Exponential Function Value** (10 points)
The exponential function f is defined by $f(x) = 11 \cdot b^x$ where b is a positive constant. If $f(4) = 2,816$, what is the value of $f(5)$?

A. 4
B. 11,264
C. 161,051
D. $164,916,224$

Answer:

□

24. **Solving for a and b** (10 points)

$$a(x+4) = b(x-1) + 1$$

If the solution to the given equation is $x = 4$, which of the following could be a value of a and b?

A. $a = 8$ and $b = -3$
B. $a = 7$ and $b = 6$
C. $a = 4$ and $b = -5$
D. $a = -4$ and $b = -11$

Answer:

□

25. **Exponential Equation** (10 points)
If $\frac{a^{2x^3}}{a^4} = a^{12}$, what is the value of 2^x?

Answer:

□

26. **Quadratic Expansion and Coefficient** (10 points)
If $(3x+a)(2x+b) = 6x^2 + cx + 6$ and $4a + 6b = 36$ for all values of x, what is the value of c?

Answer:

□

27. **Intersection of Lines** (10 points)
The graph of line q is perpendicular to the line $2x - 4y = 11$ and contains the point $(1, 6)$. Line p contains the points $\left(2\frac{1}{4}, 7\frac{1}{4}\right)$ and $\left(\frac{1}{3}, 5\frac{1}{3}\right)$. If p and q intersect at the point (a, b), what is the value of $a - b$?

A. -5
B. 1
C. 5
D. 6

Answer:

□

28. **Triangle Congruence** (10 points)
In triangles ABC and DEF, $\angle A$ and $\angle D$ each measure 48°, and the measures of $\overline{AB}$ and $\overline{DE}$ are both 5 inches. Which of the following is sufficient to prove that ABC and DEF are congruent triangles?

A. $\overline{BC}$ measures 3 inches and $\overline{EF}$ measures 4 inches.
B. $\overline{BC}$ measures 4 inches and $\overline{EF}$ measures 4 inches.
C. $\overline{AC}$ measures 4 inches and $\overline{DF}$ measures 4 inches.
D. $\overline{BC}$ measures 3 inches and $\overline{DF}$ measures 4 inches.

Answer:

□

29. **Quadratic Function Maximum** (10 points)

$$f(x) = -2x^2 + 16x - 11$$

If $g(x) = f(x-3)$, what is the value of x where $g(x)$ reaches its maximum value?

Answer:

□

30. **Algebraic Expression Simplification** (10 points)
Which expression is equivalent to $\frac{25(x^2-5)}{(x-\sqrt{5})}$, where $x \neq \sqrt{5}$?

A. $5(x-5)$
B. $5(x+5)$
C. $25(x-\sqrt{5})$
D. $25(x+\sqrt{5})$

Answer:

□

31. **Circle Radius** (10 points)
The equation below defines a circle in the xy-coordinate plane:

$$x^2 + y^2 - 6x - 8y = -16$$

What is the length of the circle's radius?

Answer:

□

32. **Arithmetic Sequence Sum** (10 points)
A sequence of numbers starts at the number a. Each sequential number is k more than the previous number. The sum of the first 13 numbers can be written as $xa + yk$. What is the value of $x + y$?

Answer:

□